

Pathology-Level ML Model Performance Summary (Test Set)
Classical ML models trained on neural network pathology predictions

Neural Network Model	Pathology	ML Algorithm	AUC Mean	AUC Std	AUC 95% CI	F1 Mean	F1 Std	Samples	Positive Rate	Cross-Val Folds
Attention Pool Extra3	A Lines	Random Forest	0.758	0.026	[0.726-0.790]	0.665	0.041	43689	0.472	5
Attention Pool Extra3	A Lines	Logistic Regression	0.795	0.014	[0.778-0.813]	0.709	0.034	43689	0.472	5
Attention Pool Extra3	A Lines	XGBoost	0.794	0.013	[0.779-0.810]	0.694	0.029	43689	0.472	5
Attention Pool Extra3	Large Consolidation	Random Forest	0.863	0.013	[0.847-0.880]	0.479	0.078	43689	0.110	5
Attention Pool Extra3	Large Consolidation	Logistic Regression	0.919	0.004	[0.913-0.924]	0.554	0.039	43689	0.110	5
Attention Pool Extra3	Large Consolidation	XGBoost	0.901	0.012	[0.886-0.915]	0.553	0.064	43689	0.110	5
Attention Pool Extra3	Pleural Effusion	Random Forest	0.674	0.058	[0.602-0.745]	0.049	0.040	43689	0.011	5
Attention Pool Extra3	Pleural Effusion	Logistic Regression	0.808	0.039	[0.759-0.856]	0.063	0.039	43689	0.011	5
Attention Pool Extra3	Pleural Effusion	XGBoost	0.829	0.041	[0.778-0.880]	0.043	0.040	43689	0.011	5
Attention Pool Extra3	Other Pathology	Random Forest	0.644	0.024	[0.614-0.674]	0.492	0.062	43689	0.405	5
Attention Pool Extra3	Other Pathology	Logistic Regression	0.690	0.022	[0.662-0.717]	0.583	0.046	43689	0.405	5
Attention Pool Extra3	Other Pathology	XGBoost	0.684	0.018	[0.662-0.707]	0.513	0.066	43689	0.405	5
Original	A Lines	Random Forest	0.704	0.036	[0.660-0.749]	0.583	0.111	43689	0.472	5
Original	A Lines	Logistic Regression	0.761	0.028	[0.725-0.796]	0.660	0.071	43689	0.472	5
Original	A Lines	XGBoost	0.745	0.035	[0.702-0.788]	0.609	0.128	43689	0.472	5
Original	Large Consolidation	Random Forest	0.812	0.042	[0.760-0.865]	0.270	0.159	43689	0.110	5
Original	Large Consolidation	Logistic Regression	0.899	0.018	[0.877-0.922]	0.413	0.073	43689	0.110	5
Original	Large Consolidation	XGBoost	0.868	0.010	[0.856-0.881]	0.294	0.157	43689	0.110	5
Original	Pleural Effusion	Random Forest	0.663	0.054	[0.596-0.730]	0.005	0.011	43689	0.011	5
Original	Pleural Effusion	Logistic Regression	0.846	0.024	[0.817-0.876]	0.069	0.027	43689	0.011	5
Original	Pleural Effusion	XGBoost	0.834	0.040	[0.785-0.883]	0.013	0.020	43689	0.011	5
Original	Other Pathology	Random Forest	0.617	0.011	[0.604-0.630]	0.464	0.035	43689	0.405	5
Original	Other Pathology	Logistic Regression	0.647	0.018	[0.625-0.669]	0.566	0.020	43689	0.405	5
Original	Other Pathology	XGBoost	0.646	0.010	[0.635-0.658]	0.471	0.055	43689	0.405	5
Original Noinitweights	A Lines	Random Forest	0.702	0.074	[0.610-0.793]	0.615	0.070	43689	0.472	5
Original Noinitweights	A Lines	Logistic Regression	0.753	0.052	[0.688-0.817]	0.686	0.036	43689	0.472	5
Original Noinitweights	A Lines	XGBoost	0.737	0.069	[0.651-0.823]	0.630	0.082	43689	0.472	5
Original Noinitweights	Large Consolidation	Random Forest	0.747	0.090	[0.635-0.858]	0.295	0.179	43689	0.110	5
Original Noinitweights	Large Consolidation	Logistic Regression	0.867	0.033	[0.826-0.908]	0.339	0.146	43689	0.110	5
Original Noinitweights	Large Consolidation	XGBoost	0.806	0.061	[0.730-0.882]	0.321	0.205	43689	0.110	5
Original Noinitweights	Pleural Effusion	Random Forest	0.613	0.049	[0.552-0.674]	0.004	0.008	43689	0.011	5
Original Noinitweights	Pleural Effusion	Logistic Regression	0.843	0.049	[0.782-0.903]	0.070	0.041	43689	0.011	5
Original Noinitweights	Pleural Effusion	XGBoost	0.785	0.046	[0.728-0.842]	0.004	0.008	43689	0.011	5
Original Noinitweights	Other Pathology	Random Forest	0.601	0.032	[0.561-0.640]	0.426	0.094	43689	0.405	5
Original Noinitweights	Other Pathology	Logistic Regression	0.653	0.035	[0.609-0.697]	0.512	0.186	43689	0.405	5
Original Noinitweights	Other Pathology	XGBoost	0.640	0.029	[0.604-0.676]	0.432	0.117	43689	0.405	5